

THE HONG KONG POLYTECHNIC UNIVERSITY
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

Subject Code : EIE4113
Subject Title : Wireless and Mobile Systems
Session : Semester 1, 2023/24 **Venue** : SH1
Date : 21 December 2023 **Time** : 12:30 – 14:30
Time Allowed : 2 Hours **Subject Examiner(s)** : Dr L Liu

This question paper has a total of 5 pages (attachments included).

Instructions to Candidates: 1. This question paper contains FOUR (4) questions.
Answer all FOUR (4) questions with clear derivations.
2. Question 1 carries 25 marks. Question 2 carries 20 marks.
Question 3 carries 30 marks. Question 4 carries 25 marks.
3. The maximum mark of this paper is 100.

Physical Constants: Nil

Other Attachments: Nil

Available from Invigilator: Nil

DO NOT TURN OVER THE PAGE UNTIL YOU ARE TOLD TO DO SO.

Q1. (25 marks)

(a) Consider the following power delay profile

$$A_c(\tau) = \frac{1}{0.000002} e^{-\tau/0.000002}, \tau \geq 0.$$

Find the mean access delay and the root mean squared (RMS) delay spread of this channel. Show how you get the results. You can rely on the following hints when you calculate the mean access delay and RMS delay spread.

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \int x e^{ax} dx = \left(\frac{x}{a} - \frac{1}{a^2} \right) e^{ax}, \int x^2 e^{ax} dx = \left(\frac{x^2}{a} - \frac{2x}{a^2} + \frac{2}{a^3} \right) e^{ax}, \lim_{x \rightarrow \infty} x e^{-x/0.000002} = 0, \lim_{x \rightarrow \infty} x^2 e^{-x/0.000002} = 0.$$

(13 marks)

(b) There are 3 senders and 1 receiver over the wireless channel, where the senders generate interference to each other. Each sender can merely transmit bit 1 or bit -1. Moreover, the interfering transmitted signals are additive. Therefore, the signal received at the receiver side is the sum of the transmitted signals from all senders.

Assume that sender 1 has a CDMA code of (1, 1, 1, -1, 1, -1, -1, -1), sender 2 has a CDMA code of (1, -1, 1, 1, 1, -1, 1, 1), and sender 3 has a CDMA code of (1, 1, 1, 1, 1, 1, 1, 1). The CDMA codes of these three senders are known by the receiver. Each sender encodes one bit, either 1 or -1, using its CDMA code and transmits its encoded signal to the receiver. Assume that the received signal at the receiver side is (1, -1, 1, -1, 1, -3, -1, -1). The receiver aims to decode the bit transmitted by each sender.

(i) Can the bit transmitted by sender 1 be decoded by the receiver? If yes, what is the decoded bit? If no, what is the reason that this bit cannot be decoded?

(4 marks)

(ii) Can the bit transmitted by sender 2 be decoded by the receiver? If yes, what is the decoded bit? If no, what is the reason that this bit cannot be decoded?

(4 marks)

(iii) Can the bit transmitted by sender 3 be decoded by the receiver? If yes, what is the decoded bit? If no, what is the reason that this bit cannot be decoded?

(4 marks)

Q2. (20 marks)

(a) Consider the code {0, 01, 011, 1110} for source coding. Is this code a uniquely decodable code? Explain your answer.

(2 marks)

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(b) Consider the code $\{0,10,101,111\}$ for source coding. Is this code a prefix code? Explain your answer.

(2 marks)

(c) Consider the code $\{0000,1011,0101,1110\}$ for channel coding. Is this code a linear code? Explain your answer.

(2 marks)

(d) Consider (7,4) Hamming code with the following generator matrix

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

and parity check matrix

$$H = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

(i) List the codewords of the following four messages: $\mathbf{m}_1=[0 \ 1 \ 1 \ 0]$, $\mathbf{m}_2=[0 \ 0 \ 1 \ 1]$, $\mathbf{m}_3=[1 \ 1 \ 1 \ 0]$, and $\mathbf{m}_4=[1 \ 0 \ 1 \ 1]$. Show how you get these codewords.

(8 marks)

(ii) Suppose a received signal is $\mathbf{r}=[0 \ 1 \ 1 \ 0 \ 0 \ 1 \ 1]$. Assume that in the received signal, there is at most 1 error bit. What is the error syndrome? What is the transmitted message?

(6 marks)

Q3. (30 marks)

(a) Consider the classful IP addresses.

(i) Show the network I.D. and the host I.D. (in binary form) of the IP address 215.162.2.1 in dotted-decimal form.

(4 marks)

(ii) Show the range of the possible IP addresses in the network 165.0.0.0 in dotted-decimal form.

(4 marks)

(b) Consider the classless IP addresses.

(i) Suppose one IP address is 158.132.152.29/19. Show the subnet mask, subnet part, and host part of the above IP address in binary form.

(6 marks)

(ii) Suppose a subnet has the following address 205.152.208.0/23. Show the range of the possible IP addresses in this subnet in dotted-decimal form.

(4 marks)

(c) Consider the network shown in the Figure 1. There are 7 routers in the network, i.e., **t**, **u**, **v**, **w**, **x**, **y**, and **z**. If there is an edge between two routers, then they are physically connected by fibers. The value assigned to each edge denotes the cost to use this edge. For example, the cost for using the path between router **x** and router **y** is 6.

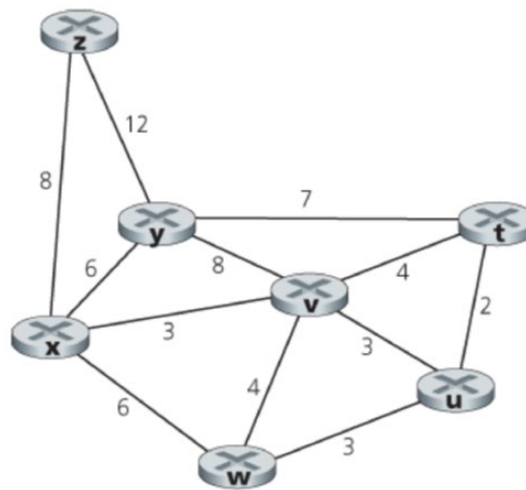


Figure 1: Routing network.

Using Dijkstra's algorithm, calculate the least-cost-path from router **u** to all the other 6 routers using the table shown below. Notations: $D(a)$: current estimate of cost of least-cost path from source to destination **a**; $p(a)$: predecessor node along path from source to **a**; N' : set of nodes whose least-cost path is definitively known.

(12 marks)

Step	N'	$D(x), p(x)$	$D(t), p(t)$	$D(v), p(v)$	$D(w), p(w)$	$D(y), p(y)$	$D(z), p(z)$
0	u						
1							
2							
3							
4							
5							
6							

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Q4. (25 marks)

(a) Describe the main difference between pure ALOHA and slotted ALOHA (your answer should be limited by 30 words).

(5 marks)

(b) Explain the hidden terminal problem in multiple access process (your answer should be limited by 30 words).

(5 marks)

(c) Explain how to utilize the RTS (request to send) and CTS (clear to send) messages to resolve the hidden terminal problem under the CSMA/CA scheme (your answer should be limited by 30 words).

(5 marks)

(d) Explain the collision issue when ALOHA is used in 5G massive machine-type communication (your answer should be limited by 30 words).

(5 marks)

(e) Suppose the permutation-based encryption technique is used due to the security issue. The function of a 15×15 permutator is given by the following table:

Input	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Output	9	8	4	6	11	15	2	14	1	5	13	7	3	10	12

If the received letters are “n s o r e e w i w r l k t s e”, then what are the transmitted letters before encryption?

(5 marks)

- End of Paper -